



DMM261 – Rule-Based Workflow – A Flexible Tool Used in SAP Master Data Governance



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DMM261 – Rule-Based Workflow: A Flexible Tool used in SAP MDG

About this session

This session provides a very practical experience of SAP Master Data Governance (MDG) and is based on the new MDG chapter in the 3rd edition of the SAP Press book "Practical Workflow for SAP". The general concept of workflow in SAP MDG and the components of the Rule-Based Workflow (RBWF) will be explained. Each attendee will build a workflow for their own change request type. With this they will be able to run the process and post a first change request to create a new material.

Agenda

Short Introduction to SAP Master Data Governance

MDG's Workflow Concept and Overview of Rule-Based Workflow

Blueprinting and Implementing a Change Request Process

Today's exercise

Short Introduction to SAP Master Data Governance

Introduction to SAP Master Data Governance

Centrally Govern Master Data for your Enterprise

SAP MDG is a state-of-the-art master data management solution, providing out-of-the-box, domain-specific master data governance to centrally create, change and distribute master data for your complete enterprise system landscape

Governance

Enables governance, compliance and transparency through integrated staging, approval and central audit trail

Consistent data

Delivers consistent definition, authorization and replication of key master data entities. Eliminates error prone manual maintenance processes for master data in multiple systems

Integration

Provide native integration and openness to integrate 3rd party services

Re-use

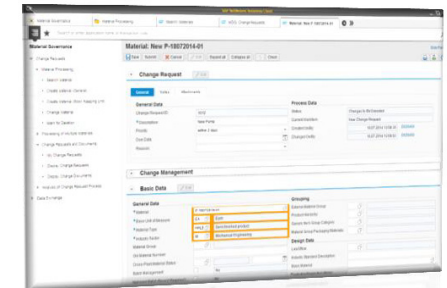
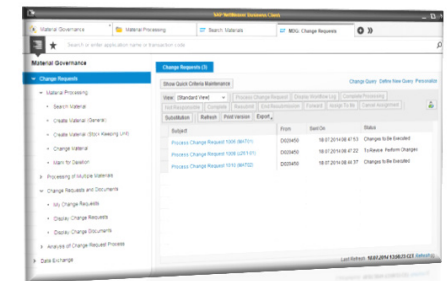
Re-use of SAP data model, existing business logic and configuration for validation of master data during creation and change

Flexibility

Open to extend the standard models and to create governance for your own master data and flexibly for non-SAP environments

Data Quality

Integrates with SAP Data Services & SAP Information Steward for quality, enrichment, and data remediation



Introduction to SAP Master Data Governance

Summary of the Key Capabilities

Master Data Governance for Financial Data

- Create or change Chart of Accounts, GL Accounts, Companies, Profit and Cost Centers, Cost Elements and respective hierarchies
- Data distribution to operational systems via Services (SOA) or ALE

Master Data Governance for Material Data

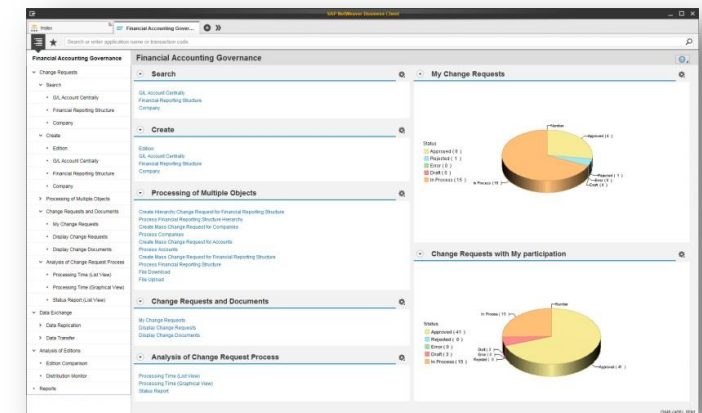
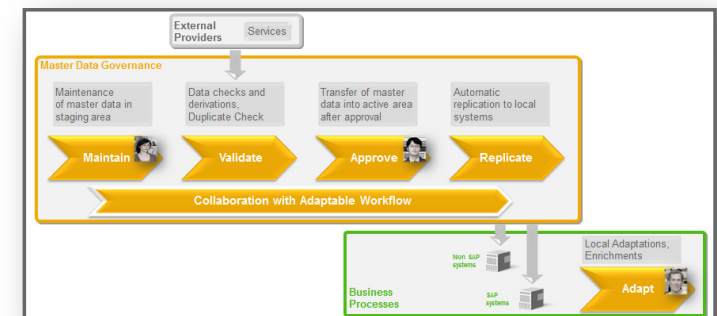
- Create or change a broad range of material master data attributes and classification
- Mass data upload and mass change capabilities
- Data quality services such as duplicate check
- Data distribution to operational systems via ALE

Master Data Governance for Supplier and Customer Data

- Create or change a broad range of supplier or customer master data attributes as well as the underlying business partner attributes
- Mass data upload and mass change capabilities
- Data quality services such as duplicate check and address validation
- Data distribution to operational systems via SOA or ALE

Master Data Governance for Custom Objects

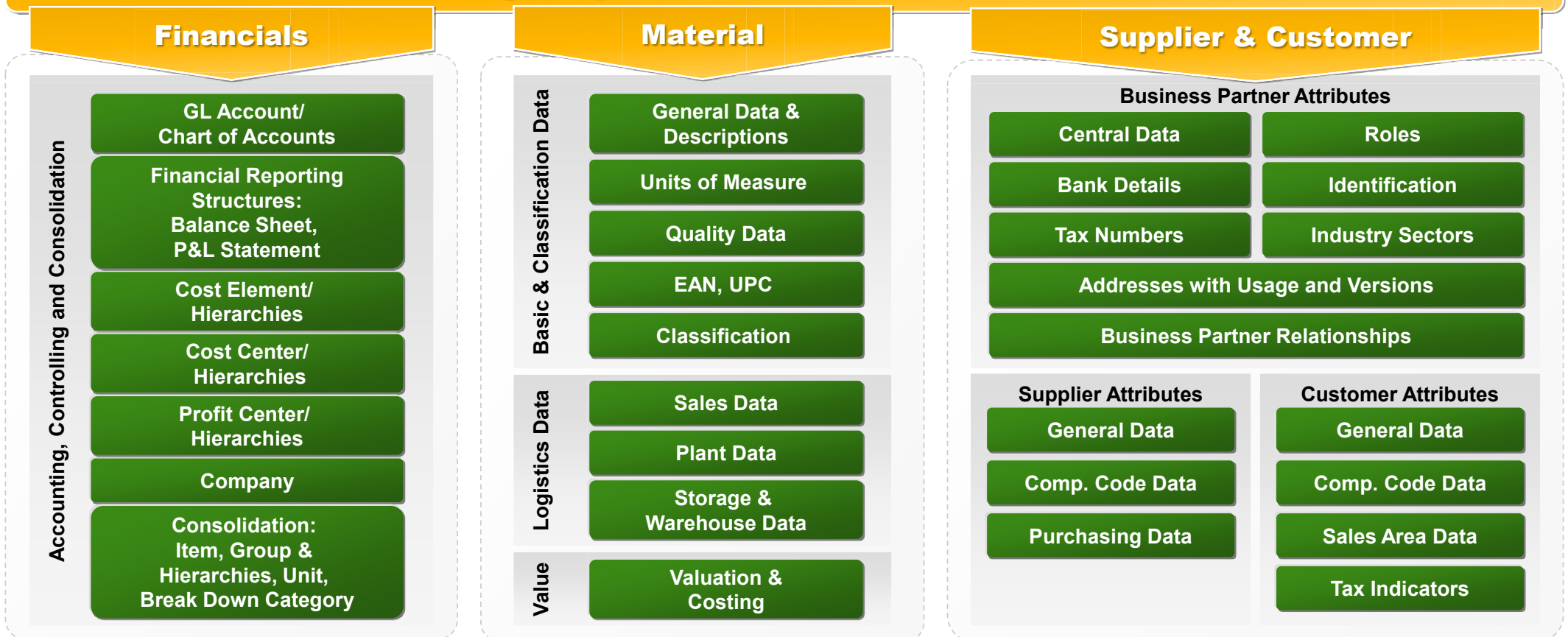
- Framework for data modeling, UI configuration, validation, and distribution of custom-defined master data objects



Introduction to SAP Master Data Governance

Out-of-the-box content provided for standard domains

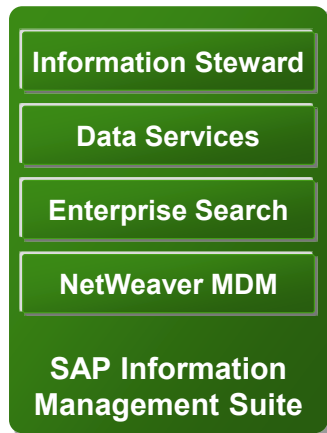
Out-of-the box delivery of proven data models for standard master data



Introduction to SAP Master Data Governance

High-level Architecture Overview

Optional Parts



SAP HANA

Central System for Master Data Processes



Portal / NetWeaver Business Client

SAP Master Data Governance

User Interfaces, Roles, Work Centers

Workflows

Financials

Business Partner

Staging

Material

Custom Objects

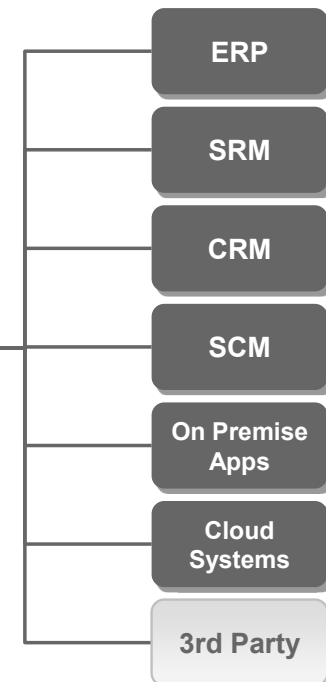
Validation

Change Request, Governance Model, Search & DQ Adapters,
UI Framework, Analytics, Replication, Key / Value Mapping

SAP ERP 6.0

AS ABAP

Client Systems



Integrate services for data cleansing & matching,
use HANA for similarity ranked search & aggregation

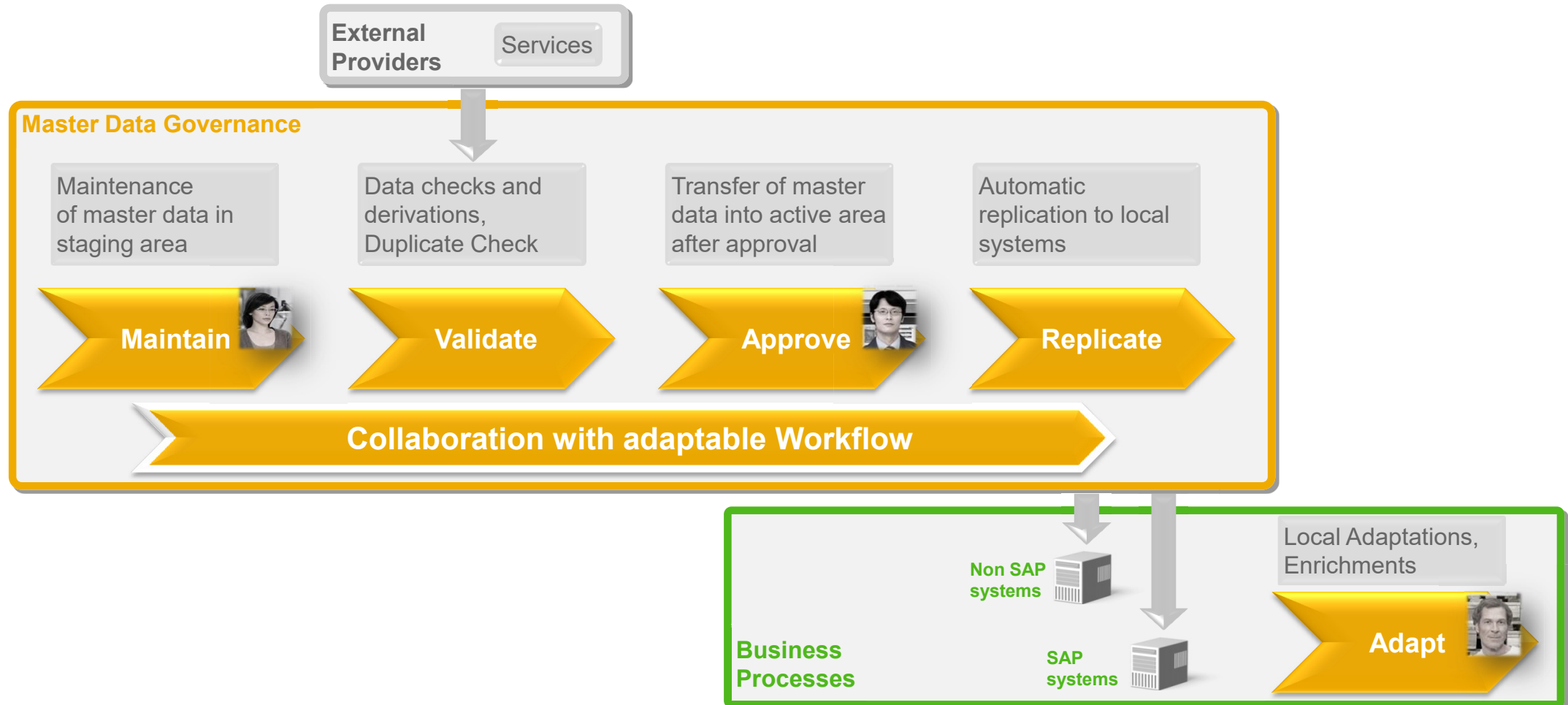
Tight integration of the governance process into SAP business
logic and leveraging standard SAP Application Server for ABAP

Client systems are integrated via various channels (like
SOA, ALE, RFC, File) across on premise and cloud

MDG's Workflow Concept and Overview of Rule-Based Workflow

MDG's Workflow Concept

Overall Process Flow

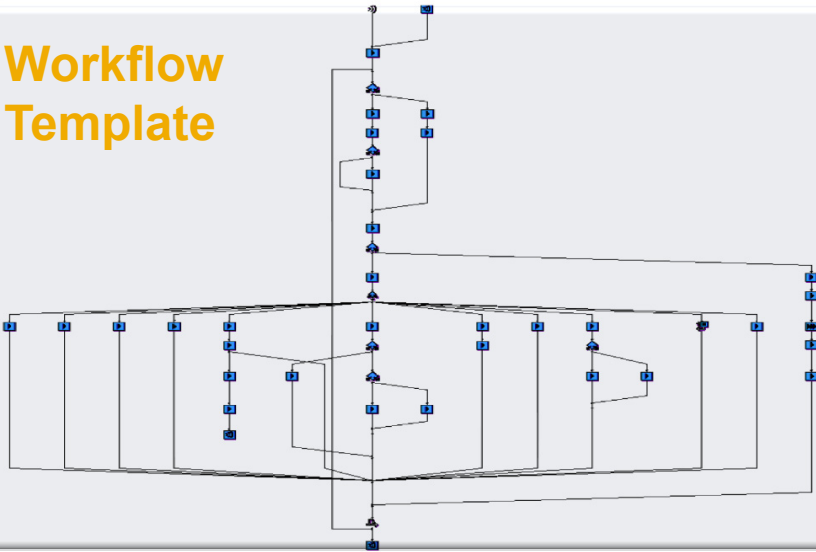


Overview of Rule-Based Workflow Components

Customizing

Type of Chg. Request	Data Model	Description (medium text)	Objects Re...	Single...	Main Entity Type	Workflow
MAT01	MM	Create Material	☐	☒	MATERIAL	WS60800086
MAT02	MM	Process Material	☐	☒	MATERIAL	WS60800086
MAT06	MM	Mark Material for Deletion	☐	☒	MATERIAL	WS60800086
MAT0A	MM	Process Multiple Materials	☐	☐	MATERIAL	WS60800086

Workflow Template



BRFplus Decision Tables

Decision Table: DT_SINGLE_VAL_MAT01

Table Contents

CR Previous Step	Previous Action	Condi...	New Chng. Req....	New CR Status
00 (Processing)		1	90 (Final Check)	02 (Changes to Be Executed)
90 (Final Check)	09 (Activate)	2	91 (Activation)	02 (Changes to Be Executed)
90 (Final Check)	10 (Send for Revision)	3	95 (Revision Pro...	02 (Changes to Be Executed)
91 (Activation)	31 (Activation successful)	4	99 (Complete)	05 (Final Check Approved)
91 (Activation)	<>31 (Activation successful)	5	90 (Final Check)	11 (Process Errors After Activation)
92 (Revision)		6	99 (Complete)	06 (Final Check Rejected)
95 (Revision Proces...	07 (Submit Request)	7	90 (Final Check)	02 (Changes to Be Executed)
95 (Revision Proces...	08 (Withdraw)	8	92 (Revision)	02 (Changes to Be Executed)

Decision Table: DT_USER_AGT_GRP_MAT01

Table Contents

Condition ...	User Agt ...	Step Type	User Agent T...	User Agent Value
1 : 5 : 7	001	5 (Activate Change Request)	US (User)	ANZEIGER
3	001	4 (Revise Change Request)	US (User)	ANZEIGER

Decision Table: DT_NON_USER_AGT_GRP_MAT01

Table Contents

Condition Al...	Agent Group	Process Pattern
2	001	05 (Activation (Do Not Bypass Snapshot))
4 : 6	001	99 (Complete (Sub-)Workflow)
8	001	08 (Roll Back Change Request)

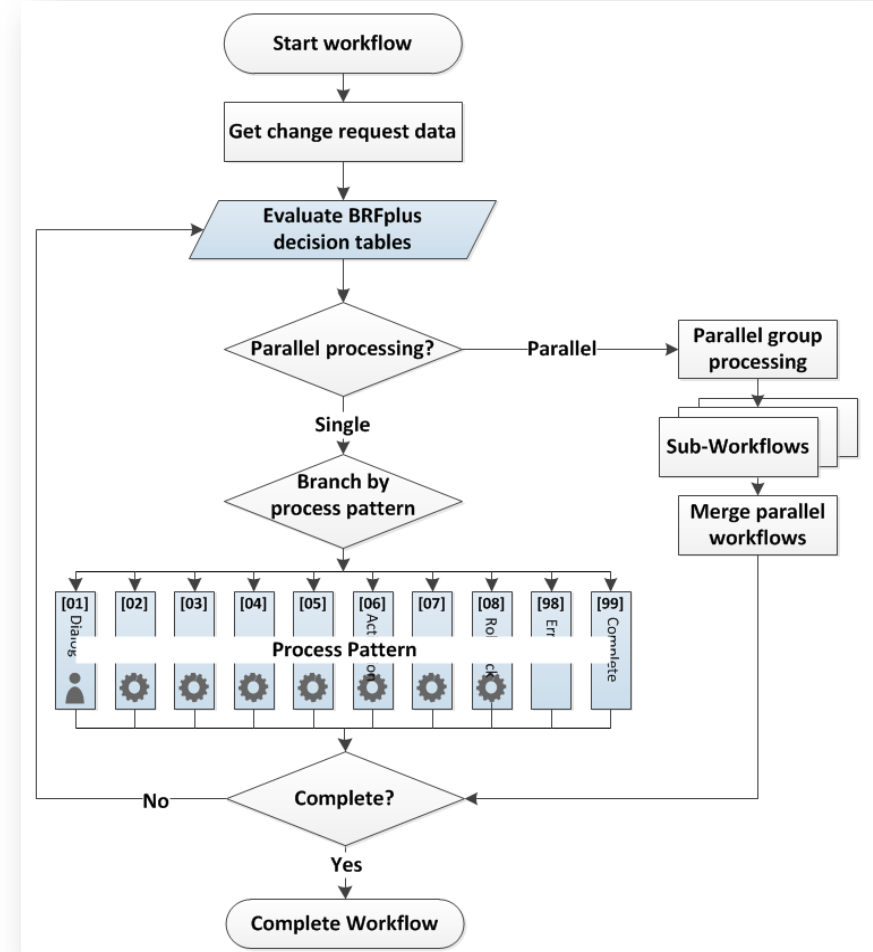
Overview of Rule-Based Workflow

Rule-Based Workflow Template

Rule-Based Workflow Template

The generic workflow template (WS60800086) is the foundation of the RBWF. The different processes are reflected by BRFplus decision tables, which are maintained in the customizing for each change request type.

The combination of this workflow template with the BRFplus decision tables allows you to easily set up many different processes that can be flexibly changed through customizing.

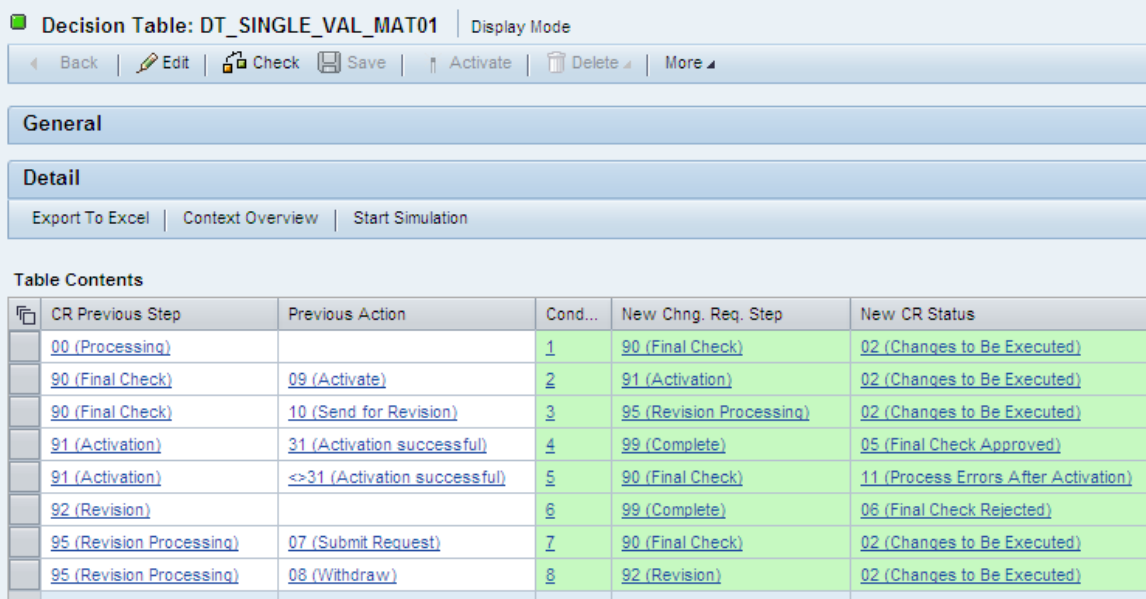


Overview of Rule-Based Workflow

Single Value Decision Table

Single Value Decision Table

The single value decision table (DT_SINGLE_VAL_<Change Request Type>) defines the flow between the change request steps. Based on the previous step, the action, and other parameters, this table returns the next step and other result parameters, most importantly, the condition alias that links to the other tables.



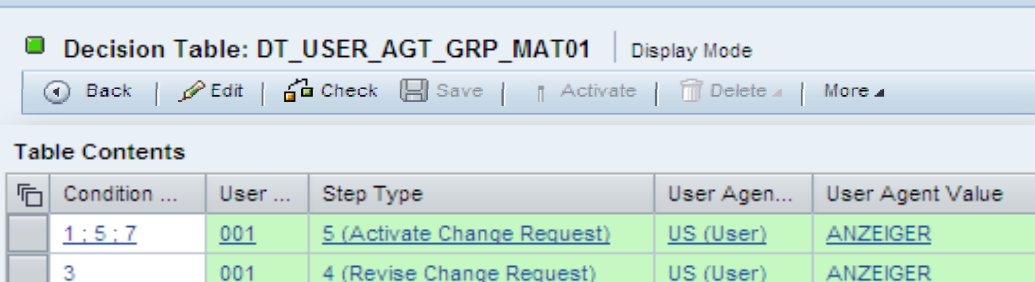
Decision Table: DT_SINGLE_VAL_MAT01 Display Mode					
◀ Back ✎ Edit 📄 Check 💾 Save ⚙️ Activate 🗑️ Delete ⌵ More					
General					
Detail					
Export To Excel Context Overview Start Simulation					
Table Contents					
CR Previous Step	Previous Action	Cond...	New Chng. Req. Step	New CR Status	
00 (Processing)		1	90 (Final Check)	02 (Changes to Be Executed)	
90 (Final Check)	09 (Activate)	2	91 (Activation)	02 (Changes to Be Executed)	
90 (Final Check)	10 (Send for Revision)	3	95 (Revision Processing)	02 (Changes to Be Executed)	
91 (Activation)	31 (Activation successful)	4	99 (Complete)	05 (Final Check Approved)	
91 (Activation)	<>31 (Activation successful)	5	90 (Final Check)	11 (Process Errors After Activation)	
92 (Revision)		6	99 (Complete)	06 (Final Check Rejected)	
95 (Revision Processing)	07 (Submit Request)	7	90 (Final Check)	02 (Changes to Be Executed)	
95 (Revision Processing)	08 (Withdraw)	8	92 (Revision)	02 (Changes to Be Executed)	

Overview of Rule-Based Workflow

User Agent and Nonuser Agent Decision Tables

User Agent Decision Table

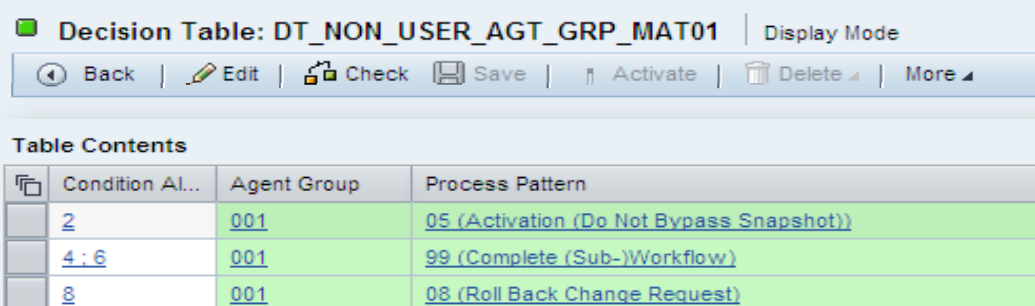
The user agent decision table (DT_USER_AGT_GRP_<Change Request Type>) determines the users involved in the change request process and which actions they can execute.



Decision Table: DT_USER_AGT_GRP_MAT01 Display Mode				
Back Edit Check Save Activate Delete More				
Table Contents				
Condition ...	User ...	Step Type	User Agen...	User Agent Value
1: 5: 7	001	5 (Activate Change Request)	US (User)	ANZEIGER
3	001	4 (Revise Change Request)	US (User)	ANZEIGER

Nonuser Agent Decision Table

The nonuser agent decision table (DT_NON_USER_AGT_GRP_<Change Request Type>) contains the background steps involved in the change request process that don't have end-user participation.



Decision Table: DT_NON_USER_AGT_GRP_MAT01 Display Mode		
Back Edit Check Save Activate Delete More		
Table Contents		
Condition Al...	Agent Group	Process Pattern
2	001	05 (Activation (Do Not Bypass Snapshot))
4: 6	001	99 (Complete (Sub-)Workflow)
8	001	08 (Roll Back Change Request)

Blueprinting and Implementing the Change Request Process

Blueprinting the Change Request Process

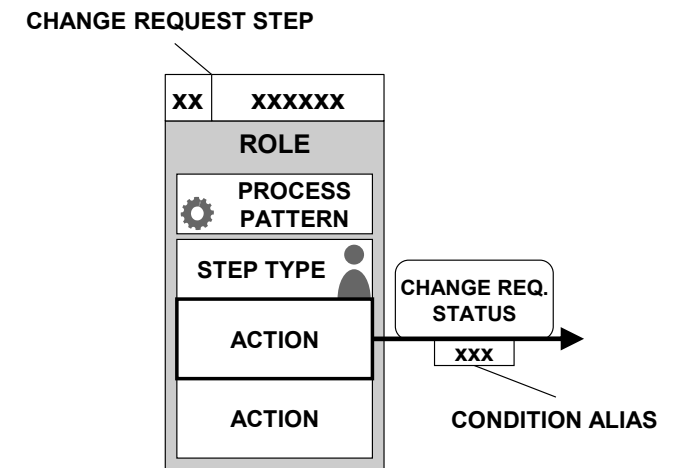
Notation for the Process Diagram

Notation

If you want to create a blueprint for the process, you must specify the following information in the process diagram:

- All steps and connections
- Change request step types or process pattern
- Change request actions
- Change request statuses
- Condition alias

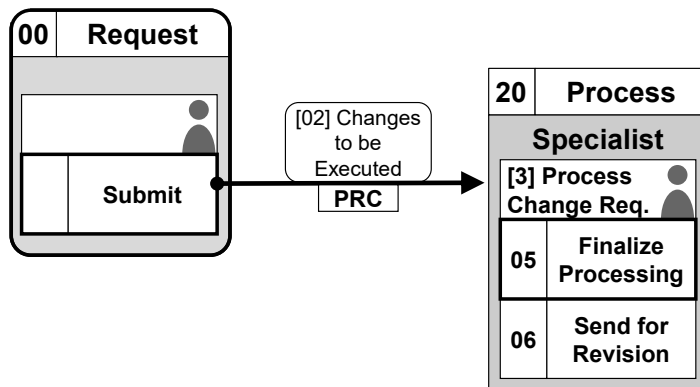
In order to create a diagram that is easy to read, we use the shown notation. This includes all relevant information needed to implement the process in SAP MDG.



Blueprinting the Change Request Process

Method with Rule-Based Workflow

- A change request process can include several approval and revision phases

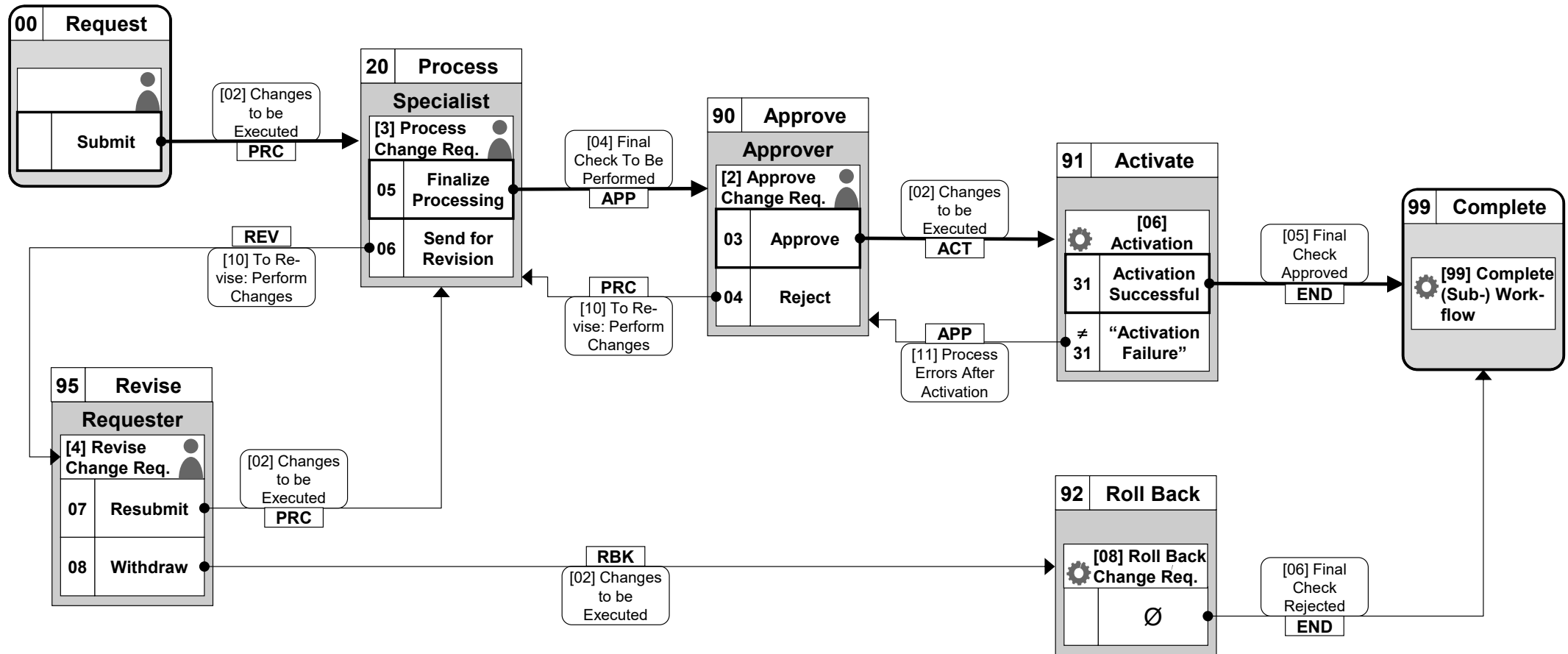


Blueprinting Your Change Request Process

- Define the process flow and decide which steps and loops are needed
- Decide which step types are needed for the dialog steps and enhance the flow diagram with out coming actions of each step
- Enhance the flow diagram with process pattern for the background steps
- Enhance the flow diagram with status
- Enhance the flow diagram with condition alias

Blueprinting the Change Request Process

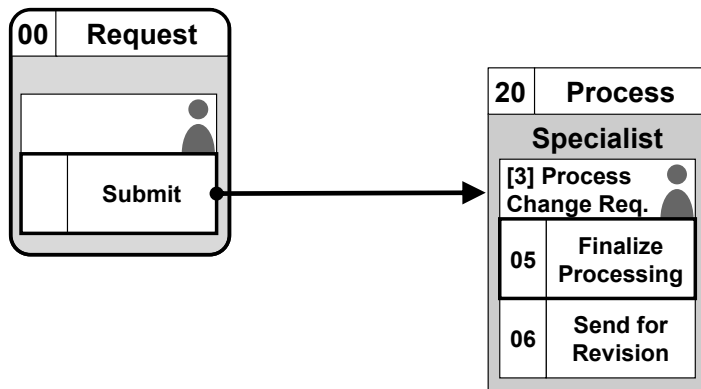
Process Diagram



Implementing the Change Request Process

Method with Rule-Based Workflow

- Rules-Based Workflow offers you a quick and easy way to build the change request process!



Implement Your Change Request Process

- Customizing: Create a New Change Request Type
- Customizing: Define Change Request Steps
- Build the decision tables
 - Single Value Decision Table
 - User Table
 - Non User Table

Implementing the Change Request Process

Customizing

Create a New Change Request Type

Change View "Type of Change Request": Overview

New Entries BC Set: Change Field Values

Dialog Structure

- Type of Change Request
 - Entity Types
 - Scope on Entity Type Level
 - Business Activities
 - Service Level Agreement for Change Request

Type of Chg. Request	Data Model	Description (medium text)	Objects Requi...	Single Object	Parallel	Main Entity Type	Workflow
Z261_150	MM	Create Material (Simple)	☐	☒	☐	MATERIAL	WS60800086
ZCONTMAS	ZS	Mass Processing of Contractors	☐	☐	☐	CONTRACTOR	WS60800086
ZCRCT...	ZS	Create Cl...	☐	☒	☐	ZRECORD	WS60800086

Define Change Request Steps

Change View "Workflow Step Numbers": Overview

New Entries BC Set: Change Field Values

Type of Chg. Request	CR Step	Keys	Validation	Description (
Z261_150	0	☐	☐	Request
Z261_150	20	☐	☐	Process
Z261_150	90	☐	☒	Approve
Z261_150	91	☐	☐	Activate
Z261_150	92	☐	☐	Roll Back
Z261_150	95	☐	☐	Revise
Z261_150	99	☐	☐	Complete

Exercise 1 – Introduction and Customizing

In this exercise you will:

- Understand the naming convention
- Logon to the system M34
- Customizing: Create a new change request type
- Customizing: Define the change request steps

Logon to the PC

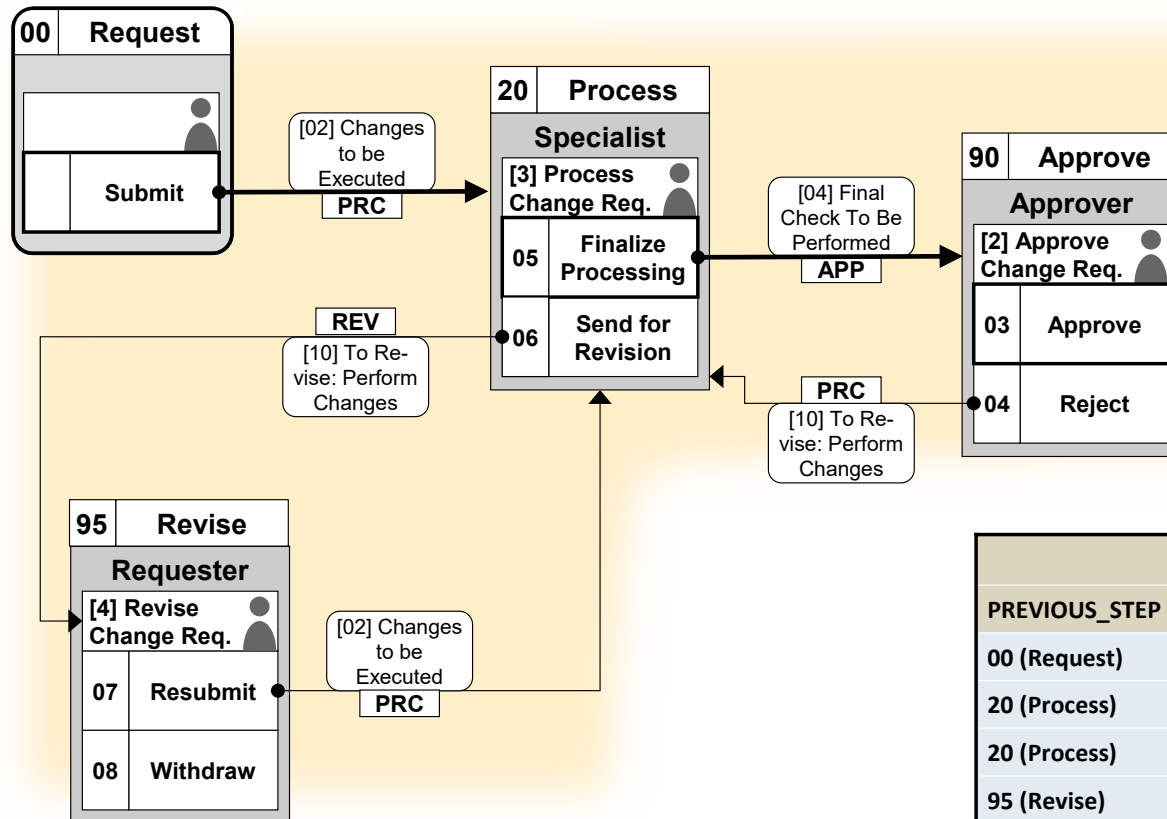
- User: Student, Domain: Fair
- Password: Welcome14



In all exercises, please make sure that you only work on **your own objects**. Here, please ensure to really pick the data reserved for your group number.

Implementing the Change Request Process

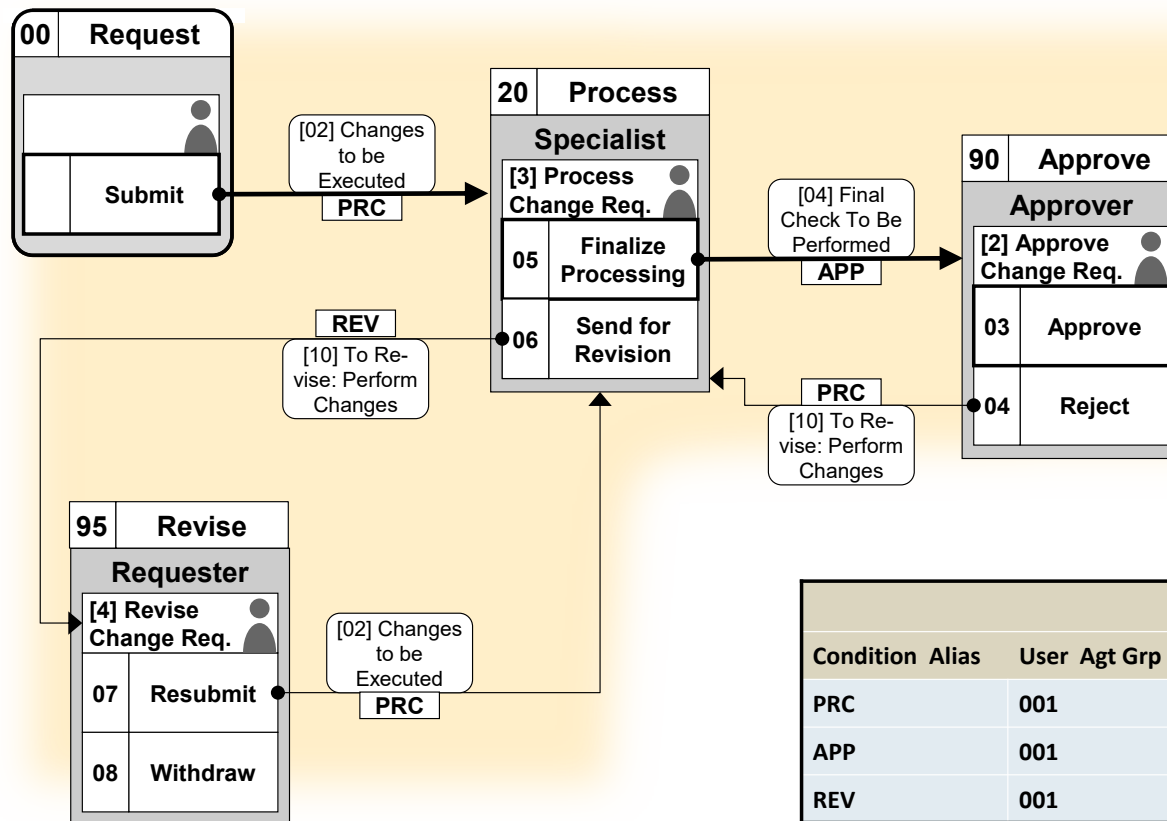
Build the Single Value Decision Table



Single Value Decision Table				
PREVIOUS_STEP	PREVIOUS_ACTION	COND_ALIAS	NEW_STEP	NEW_CR_STATUS
00 (Request)		PRC	20 (Process)	02 (Changes to Be..)
20 (Process)	05 (Finalize Processing)	APP	90 (Approve)	04 (Final Check to Be..)
20 (Process)	06 (Send for Revision)	REV	95 (Revise)	10 (To Revise: Perform.)
95 (Revise)	07 (Resubmit)	PRC	20 (Process)	02 (Changes to Be..)
95 (Revise)	08 (Withdraw)	...	...	...
90 (Approve)	03 (Approve)	...	...	...

Implementing the Change Request Process

Build the User Agent Decision Table

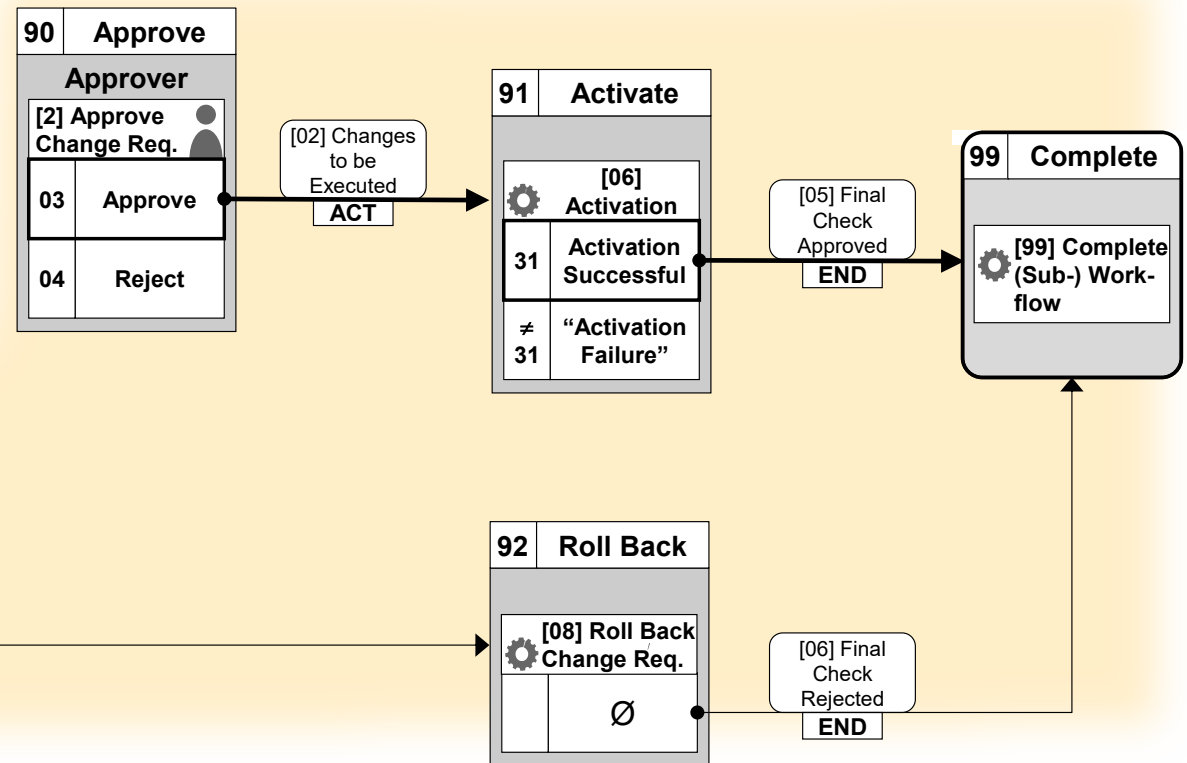


User Agent Decision Table					
Condition	Alias	User Agt Grp No.	Step Type	User Agent Type	User Agent Value
PRC		001	3 (Process Change Request)	US	Specialist
APP		001	2 (Approve Change Request)	US	Approver
REV		001	4 (Revise Change Request)	US	Requester

Implementing the Change Request Process

Build the Nonuser Agent Decision Table

Nonuser Agent Decision Table			
Condition	Alias	Agent Group	Process Pattern
ACT		001	06 (Activation)
RBK		001	08 (Roll Back)
END		001	99 (Complete (Sub-) Workflow)



Exercise 2 – Configuration

In this exercise you will:

- Maintain the single value decision table to implement the change request process, that we've blueprinted previously
- Maintain the user agent decision table
- Maintain the nonuser agent decision table
- Understand the design of the rule-based workflow built via the decision tables

In order to do so, you will execute the following steps:

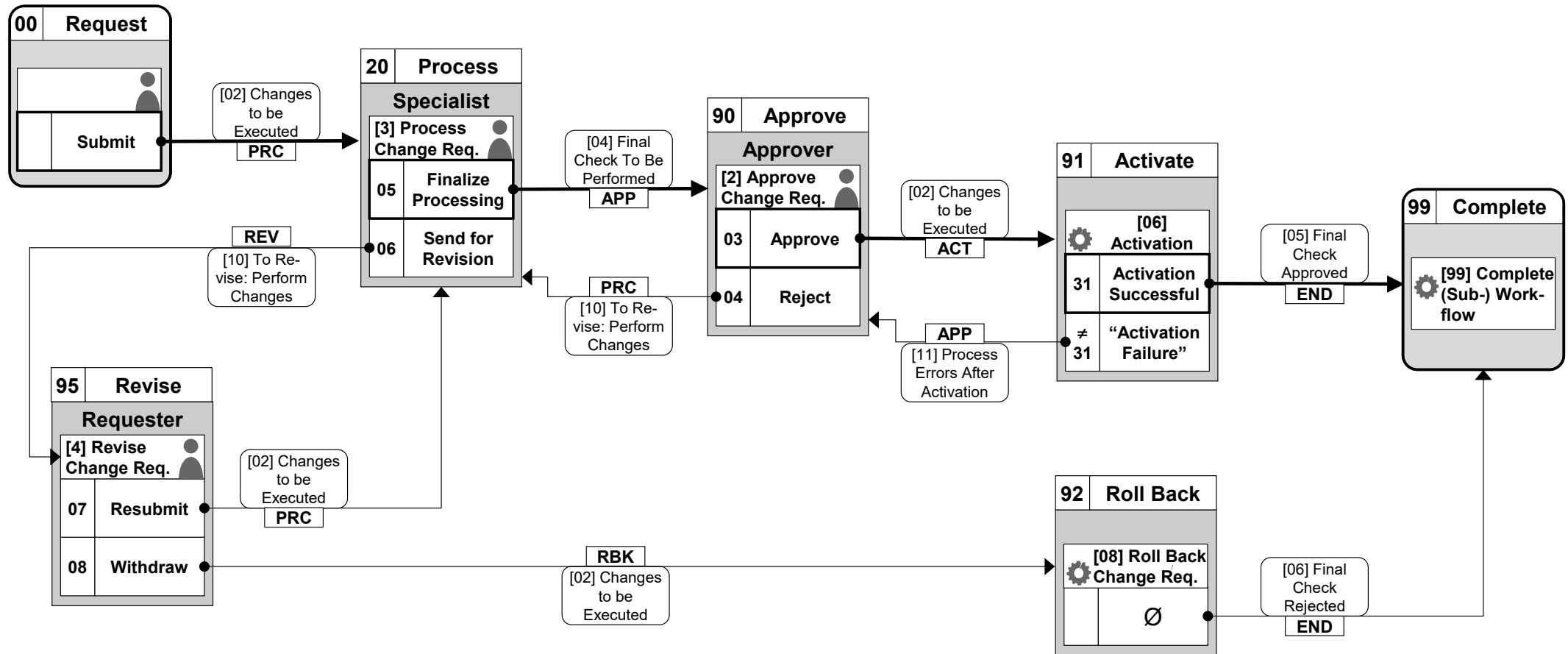
- Maintain your content in the decision tables, using the process diagram
- You can also use Transaction *USMD_SSW_RULE*.

In all exercises, please make sure that you only work on **your own objects**. Here, please ensure to really pick the data reserved for your group number.



Blueprinting Your Change Request Process

Process Diagram for the Exercise 2



Exercise 3 – Run the process

In this exercise you will:

- Logon to the system using SAP NetWeaver Business Client -Desktop (NWBC)
- Create a material using your new change request type

In order to do so, you will execute the following steps:

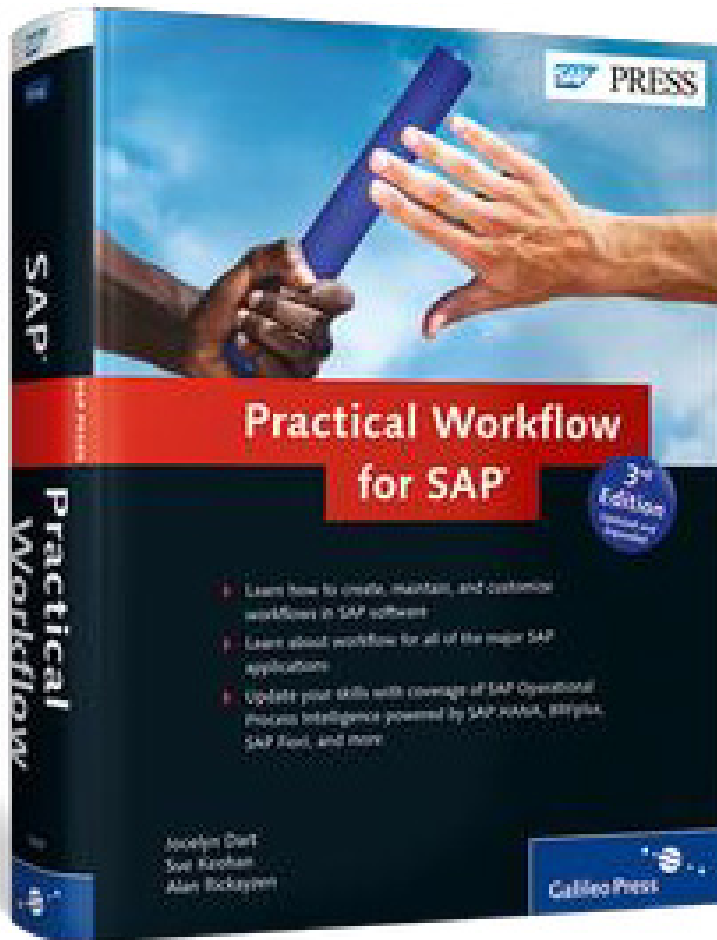
You will have 3 different roles:

- As a product manager you request a new material number
- As a product specialist you maintain additional data
- As a product approver you check and approve the data



In all exercises, please make sure that you only work on **your own objects**. Here, please ensure to really pick the data reserved for your group number.

Resources: Practical Workflow for SAP – including SAP MDG



- Get the whole story on creating, maintaining, and customizing workflows in SAP
- Learn about workflow for all the major SAP applications, including SAP Master Data Governance
- Update your skills with coverage of SAP HANA, SAP Fiori, BRF+, and more
- 3rd edition, updated and expanded
- All royalties donated to Doctors Without Borders

https://www.sap-press.com/practical-workflow-for-sap_3615/

Resources: SAP Master Data Governance SCN Space



<http://scn.sap.com/community/mdm/master-data-governance>

SAP Master Data Governance in Master Data Management (SAP MDM)

Overview



SAP Master Data Governance community homepage on SCN. SAP Master Data Governance comprises of applications to govern master data for Line of Business Processes. Used to create, maintain, and replicate master data across domains; aligned with business processes and business rules across Lines of Business (built on ECC for native SAP Business Suite integration); replication to SAP and non-SAP systems. '

Space editors:
Markus Ganser

Moderators:
Michael Theis

12-month Leaderboard

1st		Abdullah Azzouni 655 Points
2nd		Kiran Bapat 513 Points
3rd		Michael Theis 420 Points
4th		Steffen Ulmer 210 Points
5th		shankar singh 203 Points
6th		Boris Todorov 150 Points
7th		Elke Menninger 130 Points
8th		Markus Kuppe 122 Points
9th		Sanjay Shah 105 Points
10th		Markus Ganser 100 Points

Executive MDG Demos

- Financial data
- Material data
- Supplier data
- SAP Master Data Governance and SAP Information Steward

Online Help (<http://help.sap.com/mdg>)

Presentations

MDG Process, Configuration and Extensibility

- How-to Videos About MDG Processing Tasks
- Extensibility and Configuration Options for SAP Master Data Governance

SAP d-code Virtual Hands-on Workshops and SAP d-code Online

Continue your SAP d-code education after the event!

SAP d-code Virtual Hands-on Workshops

- Access hands-on workshops post-event
- Starting January 2015
- Complementary with your SAP d-code registration

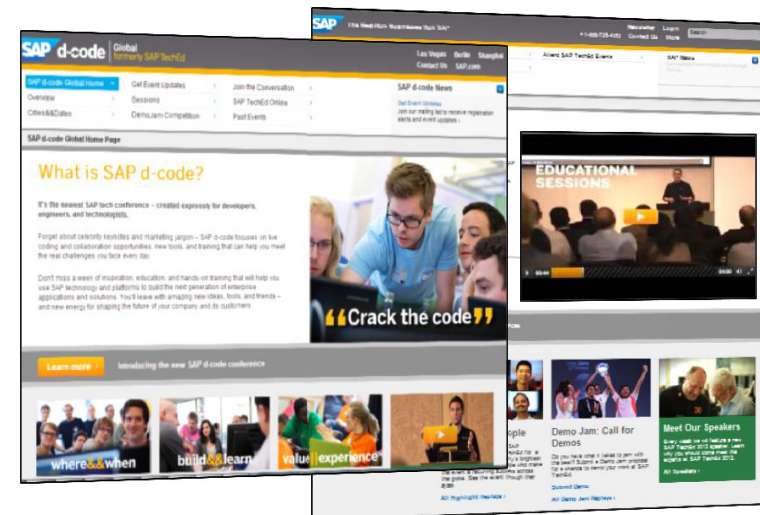
<http://sapdcodehandson.sap.com>



SAP d-code Online

- Access replays of keynotes, Demo Jam, SAP d-code live interviews, select lecture sessions, and more!
- Hands-on replays

<http://sapdcode.com/online>



Further Information

SAP Public Web

scn.sap.com/community/mdm/master-data-governance
www.sap.com/mdg

SAP Education and Certification Opportunities

www.sap.com/education
training.sap.com/search?query=master+data+governance

Watch SAP d-code Online

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Feedback

Please complete your session evaluation for
DMM261

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Thanks for attending this SAP TechEd && d-code session.



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